Find the Mistake: Composition Order and Inverse Domains

- In $(f \circ g)(x) = f\big(g(x)\big)$ the *inner* rule g acts on the input first.
- A rule is the inverse of f only if *both* round trips return the input, and only on the inputs where that actually happens.
- When a problem quotes a wrong answer, first produce the correct one, then say exactly which move produced the wrong one.

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1. Let $f(x) = 3x - 1$ and $g(x) = x^2 + 2$.

(a) Find $f\big(g(2)\big)$.

Answer: ________________

(b) Find $g\big(f(2)\big)$.

Answer: ________________

2. A student was asked for $(f \circ g)(6)$, where $f(x) = \dfrac{x}{2} + 7$ and $g(x) = 8 - x$. The student wrote -2.

(a) Work out the correct value of $(f \circ g)(6)$.

Answer: ________________

(b) Explain which composition the student actually carried out, and why it produces -2.

Answer: ________________

3. For $f(x) = 3x + 12$, a student claims the inverse rule is $\frac{x}{3} + 12$, saying "divide by 3 instead of multiplying, and leave the 12 alone."

(a) Compute $f(5)$, then feed that output into the student's claimed rule. What comes out?

Answer: ________________

(b) Write the correct rule for $f^{-1}(x)$.

Answer: ________________

4. A student writes that the inverse of $h(x) = x^2 + 1$ is $\sqrt{x-1}$, and says it works for every input.

(a) Compute $h(-3)$, then feed that output into $\sqrt{x-1}$. Write the number that comes out.

Answer: ________________

(b) Explain what the round trip shows about the claim.

Answer: ________________

5. Asked for a simplified rule for $(g \circ f)(x)$ with $f(x) = x - 4$ and $g(x) = x^2$, a student wrote $x^2 - 4$.

(a) Write the correct simplified rule for $(g \circ f)(x)$.

Answer: ________________

(b) Evaluate the correct rule at $x = 7$.

Answer: ________________

(c) Explain the error in the student's rule.

Answer: ________________

6. Let $k(x) = \sqrt{x} + 3$.

(a) Find the rule for $k^{-1}(x)$.

Answer: ________________

(b) The rule from part (a) is not the whole answer: it must carry a restriction on its inputs. State the restriction and justify it.

Answer: ________________

7. A student claims that the inverse of $f(x) = \dfrac{x+5}{x-2}$ is $\dfrac{x-2}{x+5}$ — "just turn it upside down."

(a) Find the correct rule for $f^{-1}(x)$.

Answer: __________________

(b) Evaluate your correct rule at $x = 3$.

Answer: __________________

(c) Explain why turning a rule upside down is not the same as inverting it.

Answer: __________________

8. Let $f(x) = x + 3$ and $g(x) = x^2$, both defined for every real number. One composition simplifies to $x^2 + 3$.

(a) Write the simplified rule for the *other* composition, $(g \circ f)(x)$.

Answer: ________________

(b) The two composed rules agree at exactly one input. Find it.

Answer: ________________

(c) Explain why they can agree at only one input and not at two.

Answer: ________________